1. understand the importance of chemical synthesis to provide food additives, fertilisers, dyestuffs, paints, pigments and pharmaceuticals;
2. interpret information about the sectors, scale and importance of chemical synthesis in industry and in laboratories;
3. recall the formulae of the following chemicals: chlorine gas, hydrogen gas, nitrogen gas, oxygen gas, hydrochloric acid, nitric acid, sulfuric acid, sodium hydroxide, sodium chloride, sodium carbonate, potassium chloride, magnesium oxide, magnesium hydroxide, magnesium carbonate, magnesium sulfate, calcium carbonate, calcium chloride;
4. **work out the formulae of ionic compounds given the charges on the ions (from C4);**
5. **work out the charge on one ion given the formula of a salt and the charge on the other ion (from C4);**
6. recall the main hazard symbols and understand the safety precautions to use when handling hazardous chemicals;
7. recall examples of pure acidic compounds which are solids (citric and tartaric acids), liquids (sulfuric, nitric and ethanoic acids) or gases (hydrogen chloride);
8. recall that common alkalis include the hydroxides of sodium, potassium and calcium;
9. recall the pH scale;
10. recall the use of indicators and pH meters to measure pH;
11. recall the characteristic reactions of acids that produce salts to include the reactions with metals, oxides, hydroxides and carbonates;
12. write balanced equations with state symbols to describe the characteristic reactions of acids;
13. recall that the reaction of acid with an alkali to form a salt is a neutralisation reaction;
14. **balance unbalanced symbol equations;**
15. explain that acidic compounds produce aqueous hydrogen ions, H^+(aq), in water;
16. explain that alkaline compounds produce aqueous hydroxide ions, OH^-(aq), when they dissolve in water;
17. write down the formula of the salt produced given the formulae of the acid and the alkali;
18. explain that during a neutralisation reaction, the hydrogen ions from the acid react with hydroxide ions from the alkali to make water: $H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$.